Exp No: 1.a Analyse the trend of Data Science job postings over the last decade

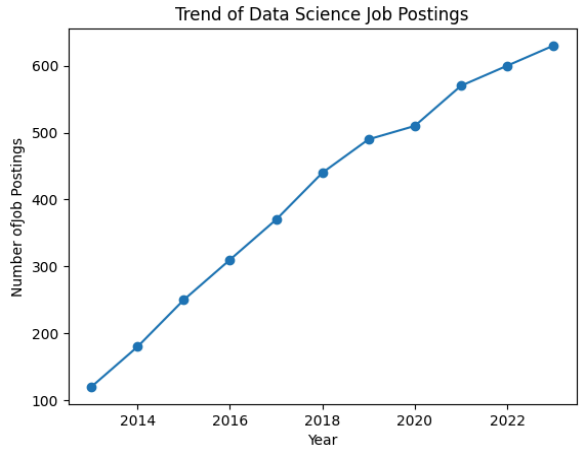
AIM:

To analyse the trend of data science job postings over the last decade.

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
data = {'Year': list(range(2013, 2024)),
'Job Postings': [120,180,250,310,370,440,490,510,570,600,630]}
df = pd.DataFrame(data)
plt.plot(df['Year'], df['Job Postings'], marker='o')
plt.title('Trend of Data Science Job Postings')
plt.xlabel('Year')
plt.ylabel('Number of Job Postings')
plt.show()
```

OUTPUT:



RESULT:

Hence the trend of job postings over the last decade has been analysed.

Exp No: 1.b Analyze and visualize the distribution of various data science roles from a dataset

AIM:

To analyze and visualize the distribution of various data science roles from a dataset.

PROGRAM:

```
roles = ['Data Analyst', 'Data Engineer', 'Data Scientist', 'ML Engineer',

'Business Analyst']

total_count = [150,350,220,660,300]

index=[5,7,9,11,13]

plt.bar(index,total_count,tick_label=roles)

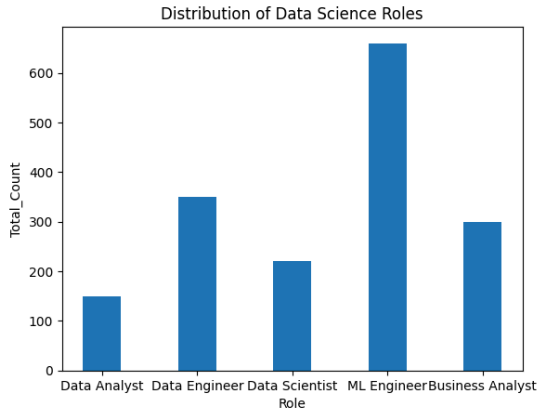
plt.title('Distribution of Data Science Roles')

plt.xlabel('Role')

plt.ylabel('Total_Count')

plt.show()
```

OUTPUT:



RESULT:

Hence the distribution of various data from a dataset is visualized and analyzed.

Exp No: 1.c Conduct an experiment to differentiate structured, un-structured and semi structured data based on data sets given.

AIM:

To conduct an experiment to differentiate structured, un-structured and semi structured data based on data sets given.

PROGRAM:

```
import pandas as pd
df=pd.DataFrame({'Roll No':[101,102,103,104],
        'Name':['Harini','Nivedha','Pooja','Poshitha'],
        'Department':['CSE','CSE','IT','RA']})
print("Structured Data:\n",df)
print("\n")
unstructured_data="Unstructured data is information that does not have a predefined data model or organizational structure."
print("Unstructured Data:\n",unstructured_data)
print("\n")
semi_structured_data = {'Roll No': [1,2,3], 'Name': ['Prithika','Harini','Berlin'], 'Attributes': {'Height': [165,155,160], 'Weight': [68,55,56]}}
print("Semi-structured Data:\n", semi_structured_data)
```

OUTPUT:

Structured Data:

	Roll No	Name	Department
0	101	Harini	CSE
1	102	Nivedha	CSE
2	103	Pooja	IT

3 104 Poshitha RA

Unstructured Data:

Unstructured data is information that does not have a predefined data model or organizational structure.

Semi-structured Data:

{'Roll No': [1, 2, 3], 'Name': ['Prithika', 'Harini', 'Berlin'], 'Attributes': {'Height': [165, 155, 160], 'Weight': [68, 55, 56]}}

RESULT:

Hence an experiment to differentiate structured, un-structured and semi structured data based on data sets given is done.

Exp No: 1.d Conduct an experiment to encrypt and decrypt given sensitive data.

AIM:

To conduct an experiment to encrypt and decrypt given sensitive data.

PROGRAM:

```
from cryptography.fernet import Fernet

key = Fernet.generate_key()

cipher = Fernet(key)

original_message = "Conduct an experiment to encrypt and decrypt given sensitive data."

data_bytes = original_message.encode()

encrypted_message = cipher.encrypt(data_bytes)

decrypted_bytes = cipher.decrypt(encrypted_message)

decrypted_message = decrypted_bytes.decode()

print("Original: ", original_message)

print("Encrypted:", encrypted_message.decode())

print("Decrypted:", decrypted_message)
```

OUTPUT:

Original: Conduct an experiment to encrypt and decrypt given sensitive data.

Encrypted:
gAAAAABqgXSYzfJkn_R9tiQ0m3ddlwJHj7RACiTWkCfTfFDltof7v9BFwDuvu-VA_bl5XINETQtoN3JCQiFPJ2DB9C33Qb3Qhd6PGR7fHxQvt6mXw3vphdmR6OA-c6Z8z

Decrypted: Conduct an experiment to encrypt and decrypt given sensitive data.

RESULT:

Hence an experiment to encrypt and decrypt given sensitive data is done.